Example ARM assembly language programs

```
; Program to add two numbers stored in registers.


     AREA    Example, CODE, READONLY  ; Define a code section named 'Example', read-only
      ENTRY                          ; Mark the entry point of the program

start   MOV     R0, #10                ; Move the immediate value 10 into register R0
      MOV     R1, #3                 ; Move the immediate value 3 into register R1
      ADD     R0, R0, R1             ; Add the value in R1 to R0, store the result in R0 (R0 = 13)

stop    B       stop                   ; Branch to 'stop' (creates an infinite loop to halt execution)

      END
```

```
; Program to add two 64-bit numbers

      AREA    Example, CODE, READONLY  ; Define a code section named 'Example',
read-only
      ENTRY                          ; Mark the entry point of the program

START    mov r0, #12  ; Add 1210 with 0508
      mov r1, #10
      mov r2, #5
      mov r3, #8

      adds r5, r1, r3 ; Add and affect flag
      adc r6, r0, r2    ; add with carry



stop    B       stop                   ; Branch to 'stop' (creates an infinite loop to halt execution)

      END                           ; End of the assembly file
```

```
;read two 16 bit numbers from memory, add them and store the result in a third memory location

        AREA Program,CODE,READONLY
           ENTRY

        LDR R0,NUM1;   get the first 16bit number from NUM1 into R0 register
        LDR R1,NUM2
        ADD R2,R0,R1
        RSB R3, R0, R1

        LDR  R4,=RESULT1
        STR  R2,[R4]
        LDR  R4,=RESULT2
        STR  R3,[R4]
        SWI  &11


NUM1    DCW   &01111
   align
NUM2    DCW  &4567
   align


   AREA DataRAM,DATA,READWRITE

RESULT1  DCD 0
RESULT2  DCD 0
          END
```

```
; ARM asm program to add from 1 to 5.

   AREA PROG2,CODE,READONLY  ; sum 1+2+3+4+5 =15 (F)
   ENTRY

   MOV R0,#0
   MOV R1,#0
BACKK ADD R0,R0,#1
   ADD R1,R1,R0
   CMP R0,#5
   BNE BACKK
GO B GO
   END
```

```
; ARM asm program to generate Fibonacci numbers from 1 to 10.

   AREA PROG13,CODE,READONLY ; Fibonacci numbers stored in memory
   ENTRY
   MOV R1,#1
   LDR R2,=TABLE
   LDR R3,=NUMFIBONACCI
   LDRB R6,[R3]
   STRB R1,[R2],#1
   MOV R3,#0
   MOV R4,#0
   MOV R5,#1
   SUB R6,R6,#1
BACKK ADD R4,R3,R1
   STRB R4,[R2],#1
   MOV R3,R1
   MOV R1,R4
   ADD R5,R5,#1
   CMP R5,R6
   BLS BACKK
GO B GO

NUMFIBONACCI DCB 0x0A
   AREA NUMBER,DATA,READWRITE
TABLE SPACE 60
   END
```

```
; ARM asm program to compute factorial of 5

        AREA    Example, CODE, READONLY  ; Define a code section named 'Example',
read-only
       ENTRY                       ; Mark the entry point of the program

       mov R0, #5
       mov R1, #1
       mov R3, #1
loop    mul R1, R3, R1
       cmp R3, R0
       add R3, R3, #1
       BNE   loop
stop    B       stop                ; Branch to 'stop' (creates an infinite loop to halt execution)

       END                         ; End of the assembly file
```